GP-800 User Manual

Catalogue

preface

Thank you very much for choosing GP-800 wireless optical probe. We hope to be your kindred spirit in future cooperation.

In order for you to use our product more conveniently, we have specially configured this user manual for you. The layout of this manual strives to be comprehensive and fast, from which you can obtain basic knowledge about installation, settings, operation, principles, etc. Please read this manual carefully before installation and use.

 The company will periodically modify the content of this manual to adapt to new features and characteristics of the product, and the changes made will be integrated into the new version. The company reserves the right to make changes to the user manual without notice.

Thank you again for using the GP-800 wireless optical probe!

Purpose

The GP-800 wireless probe solution can save 90% of auxiliary time and improve process control. Save time. Reduce the scrap rate. Maintain competitiveness.

1. Solutions for machining centers

Advanced wireless transmission mode, balancing standby time and trigger delay control. Excellent three-dimensional performance, capable of measuring three-dimensional surfaces with high accuracy, capable of channel selection, with a working distance of up to 3m (the receiving range can be increased by adapting multiple receivers); Suitable for 3D measurement of various machining centers.

2. Workpiece detection probe for turning/grinding center

3. Workpiece alignment

Using a probe can save expensive fixtures and avoid the inconvenience of manually aligning with a dial gauge. The measuring head installed on the spindle of the machining center and the tool holder of the turning center brings the following benefits:

Reduce machine downtime;

Automatic fixture, workpiece calibration, and rotation axis setting;

Eliminate manual setting errors;

Reduce waste rate;

Improve productivity and flexibility in batch product size.

The benefits of in machine workpiece inspection

The measuring heads installed on the spindle and tool holder can also be used for in-process measurement and first piece inspection - manual measurement relies on the skills of the operator, and the method of moving the workpiece to a coordinate measuring machine for inspection is often not feasible. The benefits of probe detection include:

Measure the workpiece in sequence by automatically correcting the bias value;

Enhance the reliability of unmanned processing;

Adaptive processing, providing process feedback and reducing changes;

Using automatic paranoid updates for first article detection;

Shorten the downtime waiting for the first article inspection results.

Technical indicators

Parameter	unit		Remarks
Material		304+7075	
Outer diameter of edge finder	mm	42	
Probe thread	mm	M4	
Probe length	mm	Standard configuration 30	Other sizes can be purchased on demand
Probe accuracy	mm	0.001	
Probe ball head	mm	Φ3 Ruby	
Clamping diameter	mm	8	
Clamping length	mm	25	
Repetitive accuracy	mm	0.001	1um structure is stable and reliable
XY itinerary	°	15	Exceeding the travel time will cause damage
Z itinerary	mm	2.5	Exceeding the travel time will cause damage
Wireless transmission method		2.4G wireless mode, bidirectional communication Can match multiple receivers simultaneously to increase distance	
Wireless transmission distance	m	3	Dialing chip pairing 8 probes can be turned on and used simultaneously
Probe battery	3.6V 14250 lithium battery		Standby for 3 months, power on for 1 month
Off-line working		YES	Can be used as a photoelectric splitter rod
Wireless work		YES	Can be used in conjunction with the system for automatic centering and product detection
Connection method		Wireless connection	Distance<3m
Waterproof		IP67	The tool magazine cannot be immersed in cutting fluid
Receiver voltage	V	DC 5-24v	

Structural dimensions and port definitions

Probe

Receiver

Definition of signal line color

Line color	connection	function
Black	GND	The system supplies power to the probe
Red	5-24V	
Brown	System startup io (Input low level)	Power on/off
Green	System skip (Input low level)	Trigger signal output
Yellow	System alarm io (Input low level)	Signal interruption alarm
White	System battery alarm io (Input low level)	Battery voltage alarm

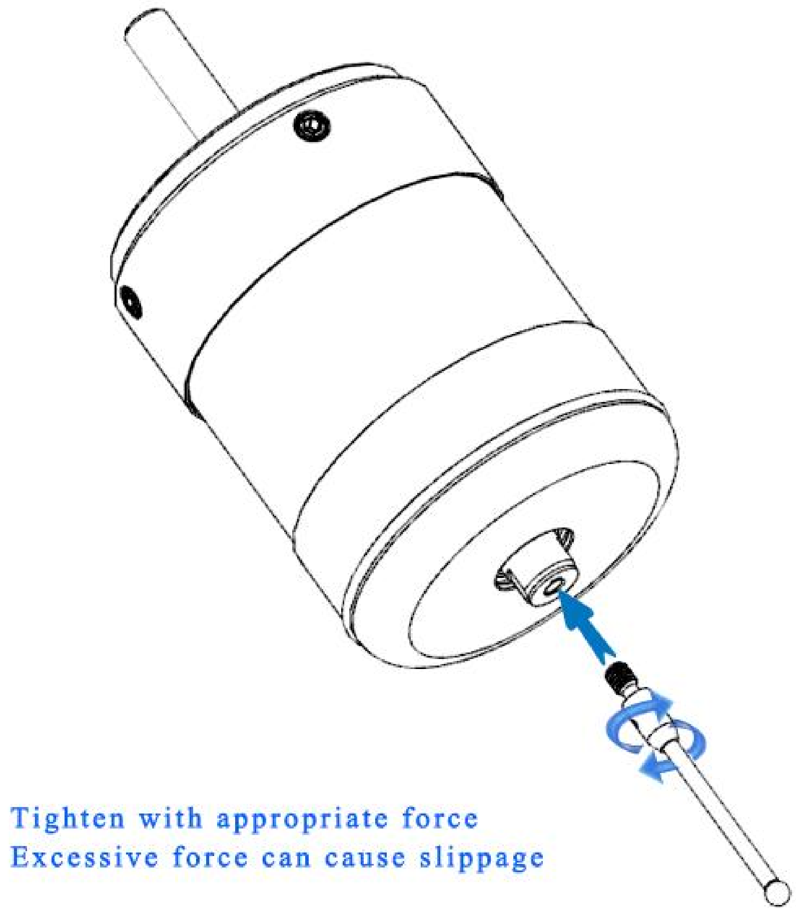
Input/output line level and on/off status

Probe power supply status			The signal is normal		Signal interruption		Low probe voltage	
	Line color	Definition	Detecting triggers	Not triggered	Power on	Shut down	Detecting triggers	Not triggered
High and low voltage levels and states	Black(input)	GND	0	0	0	0	0	0
	Red (input)	5-24V	1	1	1	1	1	1
	Brown(input)	Power on	0	0	0	–	0	0
	Green(output)	Skip	0	1	1	1	0	1
	Yellow(output)	Alarm	1	1	0	1	1	1
	White (output)	Low voltage	1	1	1	1	0	0
Explanation: 0-Low Level 1-High Level – Suspended (Input/Output Relative to Receiver)								

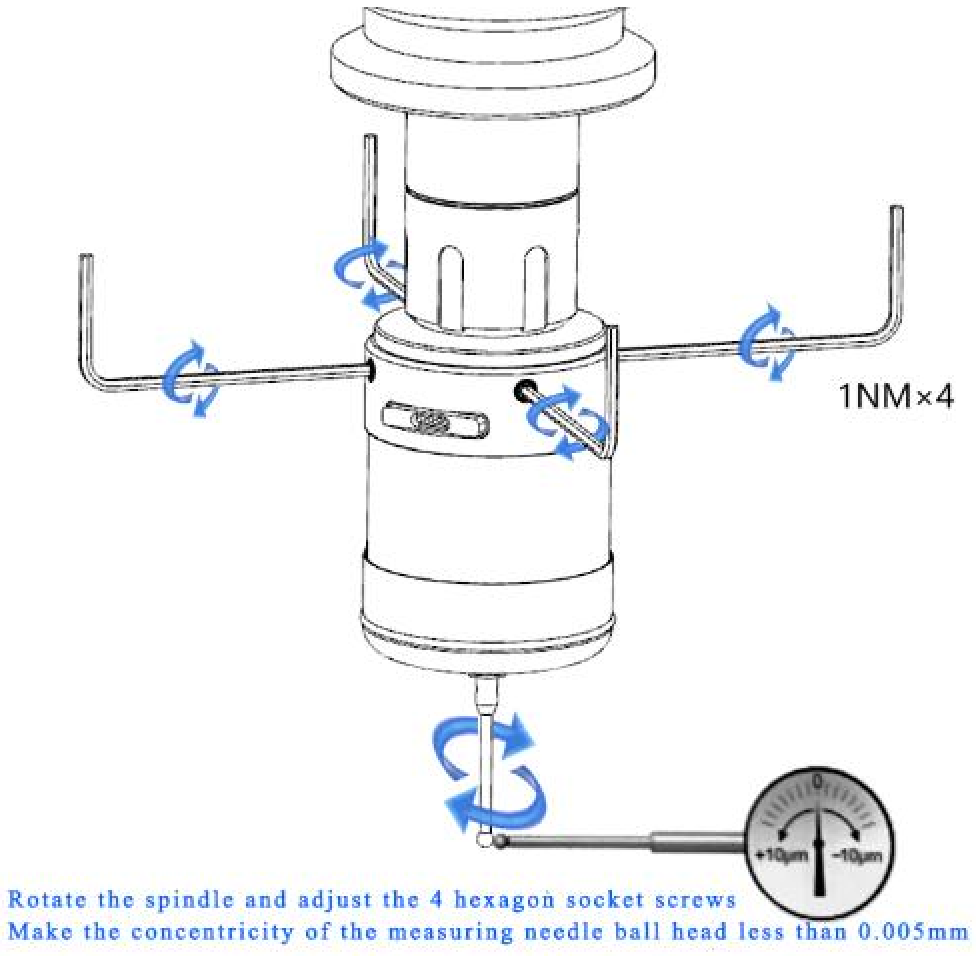
If other levels are required, the triggering mode can be set through the system, or you can purchase a high and low level conversion module yourself

Install

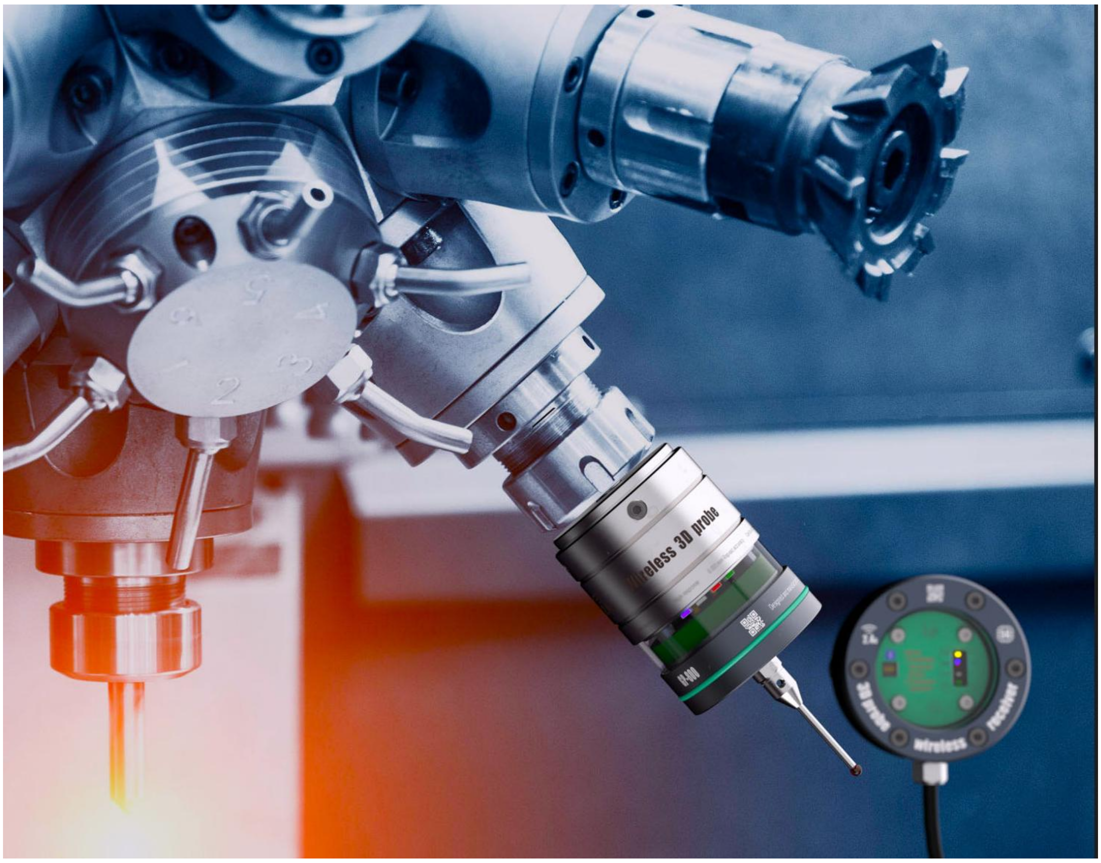
Installation of measuring needle



Zero adjustment of measuring head



Relative position of probe and receiver



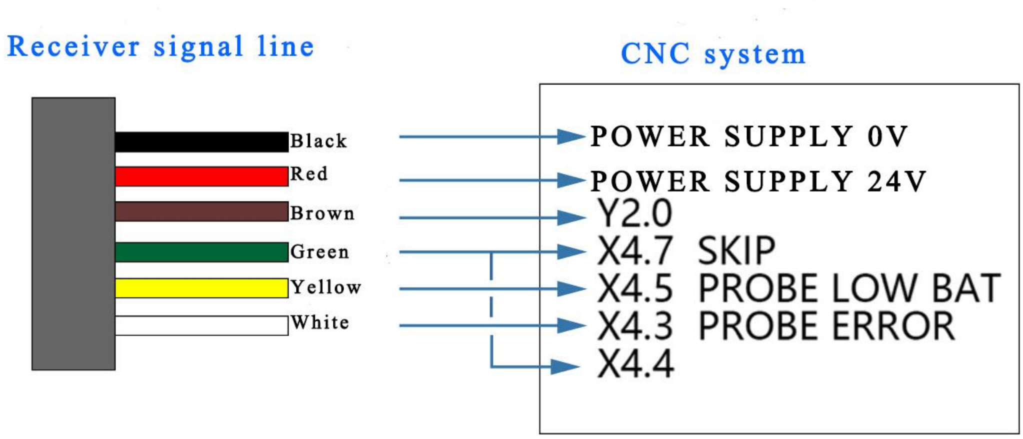
In order to maintain signal stability, the distance between the receiver and the probe should be less than 3m. Multiple receivers can be installed for long-distance relay

Wiring Example

FANUC-31i installation of GP-800 wireless probe

Wiring diagram

Pay attention to the high and low levels of system requirements. The following wiring has not been considered, only the location



The machine parameter setting X4.4 should be short circuited with X4.7

1. K17. 2=1
Effective control of contact probe
K23. 0=1
Contact probe
K32. 0=1
When detecting anomalies between M38-M39, an alarm will be triggered
6200 # 7 SKF=1, the rest are 0
6202 # 0=1, the rest are 0
2. Signal allocation for external interfaces:
Set Y2. 0 as probe open,
X4. 7 is set as a jump,
X4. 5 is set as the probe battery drops,
X4. 4 is set to the probe state,
X4. 3 Set as probe error 4. Probe command switch:
3. M17: Open probe, M18: Close probe

Programming Examples

Standard macro program

Programming Tutorial

NOTE:

1. In the following text, please use M17 M18 for the rotation opening and closing method.
2. There may be certain differences in system settings and programming methods, please pay attention to the safety of operators and machine tools.
3. The following programming tutorial is for reference only and may contain bugs.

一. Calibration of GP-800 probe: Probe calibration is to correct the deviation of the probe ball relative to the spindle centerline, the length error of the probe, and the radius error of the probe ball.

1 Calibration of the probe is required in the following situations:

A. When using the probe for the first time。

B. A new probe has been installed on the probe。

C. When the probe is suspected to be bent or the measuring head collides.。

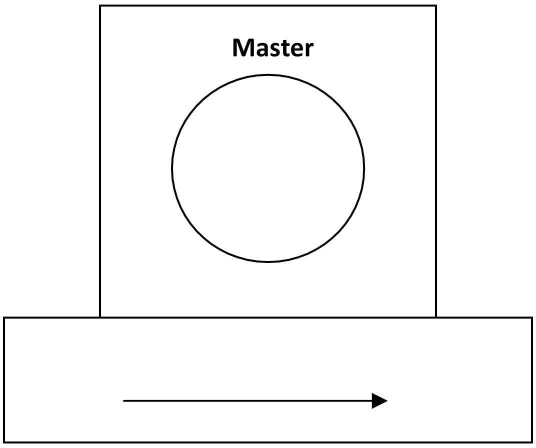
D. Periodically calibrate to compensate for mechanical variation errors of the machine tool.

E. If the repeatability of the repositioning of the probe handle is poor.。

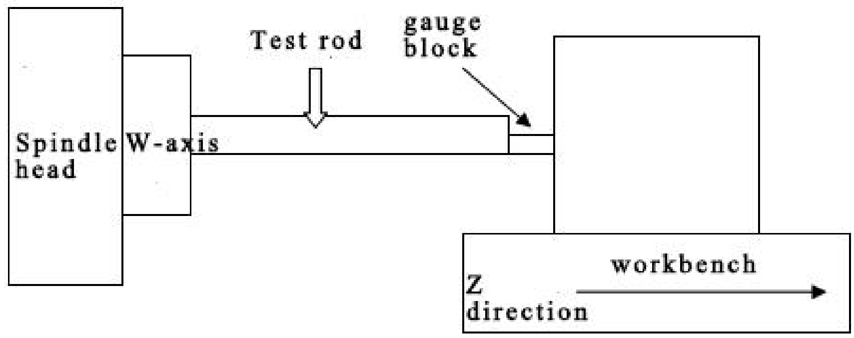
2 Place the Master (a standard block used for calibration along with the measuring head) with the known inner hole diameter on the workbench and on the side close to the spindle.

a. As shown in Figure 1, use a dial gauge to Master

(Figure 1)

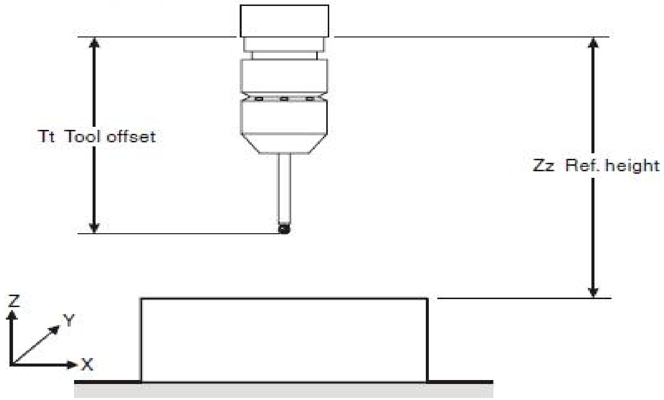


Flatten along the X direction and fix horizontally on the tabletop;
b. Use a dial gauge to find the center coordinate position of the inner hole on the Master (place it in G54X-Y -);
c. Install a Testbar on the spindle and move it
Measure the position of the master on the Z-axis using a block gauge, as shown in Figure 2
d. Place the w axis at the origin position, for example
Testbar length=350.311mm
Block gauge length=30.00mm
At this time, the Z-axis mechanical coordinate is -1148.291mm
(Figure 2)



e. The workpiece coordinate system used for setting calibration is Z=-1148.291+(-30.0)+(-350.311)
=-1528.602mm (place it on G54Z -- -)
f. Execute T1M06 (because T01 is pre-set as a probe specific);
g. Install the measuring head onto the spindle, wipe the measuring ball clean, and use a dial gauge to measure the jumping of the measuring ball. If the jumping is too large, it needs to be readjusted (there are adjustment screws on the measuring head handle in all four directions);
h. The measuring head must be installed in the spindle hole consistently and cannot be rotated 180 degrees before installation to avoid errors;
3. Complete calibration of the probe requires 09801, 09802, and their 09803 or 09804 program calibration procedures to be described in sequence:
1. (09801) Calibration of probe length:

Calibrating the probe length (O9801)



Explain

Position the side head near the Z-axis reference plane used for calibration. After the cycle is completed, the actual tool offset of the measuring head is adjusted based on the position of this reference plane.

Application

First, call the tool offset value approximated by the probe, and then position the probe close to this reference plane. Run this loop, measure the plane and reset the tool offset to a new value. After the loop is complete, return the measuring head to the starting position.

The format is as follows:

G65P9801Zz Tt;

Example: Set the values of X, Y, and Z in the G54 workpiece coordinate system;

O0001 G90G80G40G0 G54X0Y0

G43H01Z100. (As the probe is usually set to T01, activate the No.1 correction and position it at 100mm) G65M17 (opening the probe includes spindle positioning)

G65P9810Z10. F3000 (protecting positioning and moving)

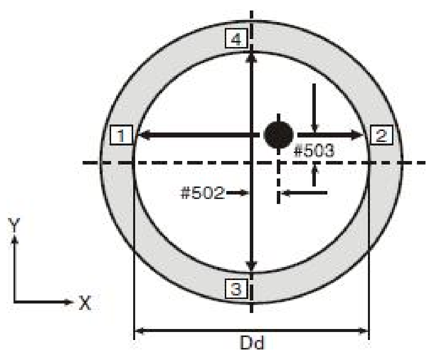
G65P9801Z0T1 (Z-direction calibration, T1 represents tool repair number)
G65P9810Z100 (protection positioning moved to Z100.0)

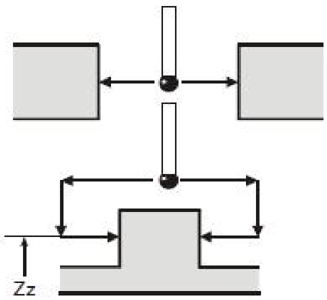
G65M18 (turn off probe)

G28Z100. (Reference point return)

H00 (Cancel Knife Repair) M30

2 (O9802) Calibration probe's X and Y direction eccentricity:





Format:

G65P9802Dd Zz

The above D represents the inner diameter value of the standard block. When using a convex platform for calibration, Zz should b

#502 = X-axis stylus offset

#503 = Y-axis stylus offset

Example 2: X Y eccentricity of calibration probe (alternative method)
The complete positioning and calibration program is as follows:

Set the accurate X, Y, and Z positions of the profile to a coordinate system (using G54 in the example)

```
 00002
G90G80G4
0G0
G54X0Y0
G43H1Z100
.
G65M17(ON)
G65P9810Z-5.0F30
00 G65P9802D50.0
G65P9810Z100.0F3
000 G65M18(OFF)
G28Z100.0
H
0
0

M
3
0
```

3 Calibrate the radius of the probe ball (09803):

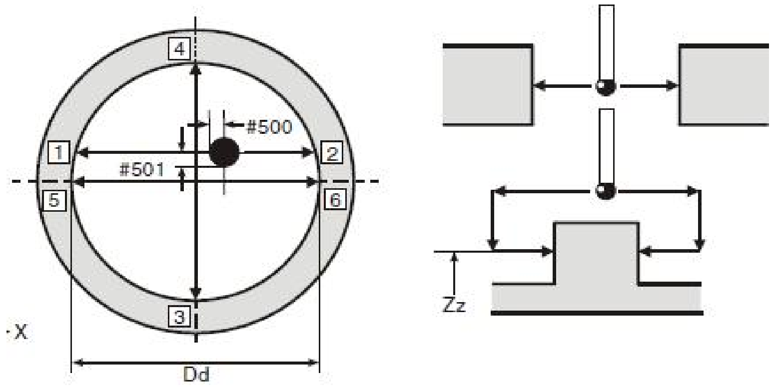


Figure 6.3 Calibrating the stylus ball radius

Application

Firstly, fix a ring gauge to an approximate known position on the machine tool worktable. With effective spindle orientation, position the measuring head to be calibrated near the center inside the ring gauge.

When running this loop, perform six movements to determine the radius value of the probe. After the cycle is completed, the probe returns to the starting position.

format:G65P9803Dd{Zz Ss}

Remarks:

D: Indicate the inner diameter size of the ring gauge;

Z: The absolute measurement position of the Z-axis is calibrated with an outer circle, and when not set, it is calibrated with a ring gauge

S: The workpiece coordinate systems S1 to S6 represent G54 to G59, while S101 to S148 represent G54.1P1 to

G54.1P48

The measured data is stored in the following macro variables:

#500=X+, X -, probe ball radius #501=Y+, Y -,

Example 1: Calibrate the radius of the probe ball

Before using this program, a tool offset must be activated. If the machine tool does not retain this bias, use an example of an alternative method.

Roughly position the probe at the center of the ring gauge. And located at the required depth

The radius of the probe ball

00003
G90G80G40G0
0
G65M17 (Rotate to open the probe)
G65P9803D50.001 (Calibrate within a ring gauge with an inner diameter of 50.001) G65M18 (Close the probe)
M30

Example 2: Calibrate the radius of the probe ball (alternative method)

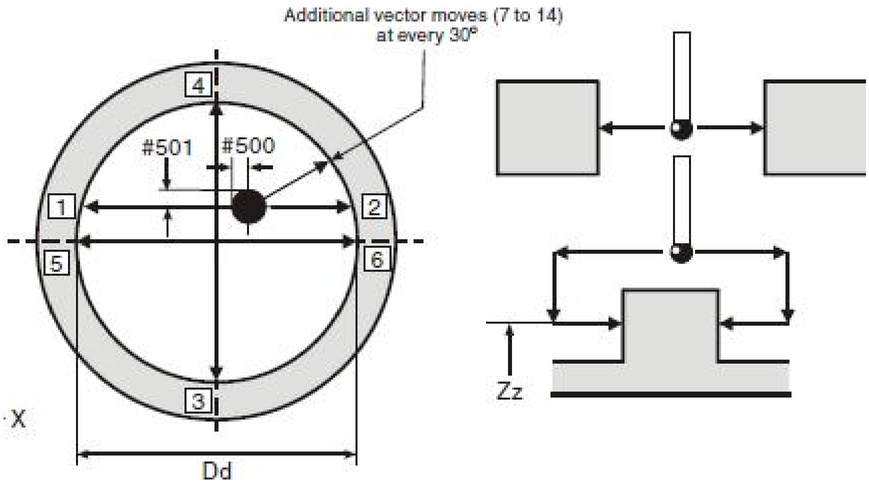
The complete positioning and calibration program is as follows:

Set X. for surface approximation Y. Z position to a workpiece coordinate system (G54 is used in the example)

00003
G90G80G40G
00 G54X0Y0
G43H1Z100.
G65M17 (Open the probe)
G65P9810Z-5.F3000 (Protective positioning to -5.0 inside the hole)

G65P9803D50.001
(Calibrate with a ring gauge with an internal diameter of 50.001)
G65P9810Z100.F3000(Protective positioning with an absolute position of 100.0)
G65M18 (Close the probe)
G28
Z10
0.
H00
M30

6. （09804）Calibrate the vector radius of the probe ball



Note: If you want to use the vector measurement subroutines 09821, 09822, or 09823, you must use this

Cycle to calibrate the radius of the probe ball. Do not use the 09803 program for calibration

Explain

Position the probe to the appropriate calibrated depth within the calibrated ring gauge. After the loop ends, the radius value of the probe ball will be saved. We obtained calibration results for 12 directions with a radius interval of 30 degrees.

Application

Firstly, fix a ring gauge to an approximate known position on the machine tool worktable. When the spindle orientation is effective, position the measuring head to be calibrated near the center of the ring gauge.

When running this loop. Perform 14 movements to determine the radius value of the probe. After the cycle is completed, the probe returns to the starting position.

Format

G65P9804Dd[Zz Ss]

remarks:

D: The size of the calibrated ring gauge;

S: Workpiece coordinate system S1~S6 (G54~G59) S101~S148 (G54.1P1~G54.1P48)

Z: When using an outer circle to represent the position of the Z-axis,

Example 1: Calibration of vector probe sphere radius

Before using this program, a tool offset must be activated. If the machine tool does not retain this bias. An example of using alternative methods.

Roughly position the probe at the center of the ring gauge and at the desired depth.

do not omit it when calibrating the ring gauge;

```
O0004
G65M17                    (Open the probe)
G65P9804D50.001(Calibrate the ring gauge with an internal diameter of 50.001)
G65M18                    (Close the probe) M30
```

O0003

Example 2: Calibration of vector probe sphere radius (alternative method)

The complete positioning and calibration program is as follows:

Set X. for surface approximation Y. Z position to a workpiece coordinate system (example using G54)

```
G90G80G40G0
0 G54X0Y0
G43H1Z100.
 G65M17                     (Open the probe)
 G65P9810Z-5.F3000 (Protective positioning
 to -5.0 inside the hole)
 G65P9804D50.001 (Calibrate with a ring
 gauge with an internal diameter of 50.001)
 G65P9810Z100.F3000(Protective positioning
 with an absolute position of 100.0)
 G65M18                     (Close the probe)
 G28
 Z10
 0.
 H00
 M30
```

《The above calibration can be completed using any of the following programs in combination》

a. Calibration methods using inner holes and surfaces

This example uses a ring gauge with a diameter of 50.001mm (1.9685in) and a known center position and end height.

Before running this program, an approximate probe length must be saved in the tool bias register. Set the X. for surface preparation Y. Z position to a workpiece coordinate system

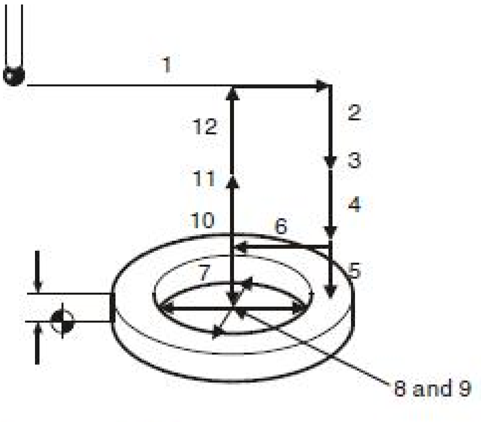
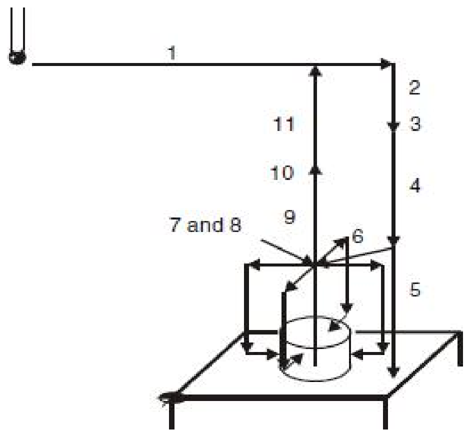


Figure 6.5 Full calibration in an internal feature

(example using G54)

O0006;
0.G90 G80 G40 G0
1. G54 X35.Y0
2 G43H01Z100.0
3.G65M17 (Open the probe)
4.G65P9810Z30.F3000 (Protect moving above the reference plane)
5.G65P9801Z20.00T1 (Calibration probe length)
6. G65P9810X0Y0 (Protect moving to the center of the hole)
7. G65P9810Z5. (Protect moving into the hole)
8. G65P9802D50. (Calibrate with a hole diameter of 50mm)
9.G65P9804D50.001 (Calibrate with a hole diameter of 50.001mm)
10. G65P9810Z100.F3000 (Protection returns to above the reference plane)
11. G65M18 (Close the probe) 12.G28Z100.
H
0
0

M
3
0
a.Calibration method using outer circle and surface



This example uses a race gauge with a known center position and Z-reference plane, and a diameter of 50.001mm (1.9685in).

Before running this program, an approximate probe length must be saved in the tool bias register. Set the X and Y center positions of the plug gauge and the zero point of the Z plane to a workpiece coordinate system (G54 is used in the example)

O0006

	G90G80G40G0	Preparatory codes for the machine.
1.	G54X135.Y100.	Move to centre of feature for height setting.
2.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in) above.
3.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
4.	G65P9810Z30.F3000	Protected positioning move above reference surface
5.	G65P9801Z0.T1	Calibrate the probe length. Z surface at zero.
6.	G65P9810X100.Y100.	Protected positioning move to centre.
7.	G65P9802D50.001Z10.	Calibrate on a 50.001 mm (1.9685 in) diameter pin gauge to establish the X,Y stylus offset.
8.	G65P9804D50.001Z10.	Calibrate on a 50.001 mm (1.9685 in) diameter pin gauge to establish the ball radius values, including the vector directions.
9.	G65P9810Z100.F3000	Protected positioning move retract to 100 mm (3.94 in).
10.	G65P9833	Spin the probe off (when applicable).
11.	G28Z100.	Reference return.
	H00	Cancel offset (when applicable)
	M30	End of program

For example, you can also write your own program to calibrate it

O0500(BLOCK CARRY) G90G80G40G0

G54X15.Y0W0B0 G43H1Z100.

M52 (Probe open)

G4X1. S100 M19

G65P9810Z30.F1000(Protective positioning and mobility)

G65P9801Z0T1 (Calibration of probe length, Z is the reference surface position, T1 is the activated No.1 tool compensation)

G65P9810X0Y0 (Locate to X0Y0 position) G65P9810Z-5. (Locate to the position of Z-5.0)

G65P9802D20.002 (The eccentricity of the calibration probe in the X and Y directions, where D is the aperture of the calibration object) G65P9804D20.002 (Calibrate the vector radius of the probe ball, where D is the aperture of the calibration object)
/G65P9830K13. (K13 multi probe probe number, K different measurement results are stored in different locations)
G65P9810Z100. (Locate at Z100 position)
G00G49Z300.
M50

二.Measurement section

1. Program Description:

a. Probe calibration program

09801 :Calibration probe length;
09802 :Calibrate the X and Y eccentricity of the probe;
09803 :Calibrate the radius of the probe ball(09821~9823 Not available)
09804 :Calibrate the vector radius of the probe ball;

b. Measurement cycle program

09810:Protective positioning (i.e. the machine will stop when the probe touches during movement);
09811:XYZ single plane measurement;
09812:Convex/groove measurement
09814:Inner hole/outer circle measurement;
09815:Inner corner measurement;
09816:Measurement of outer corners;

c. Vector measurement cycle

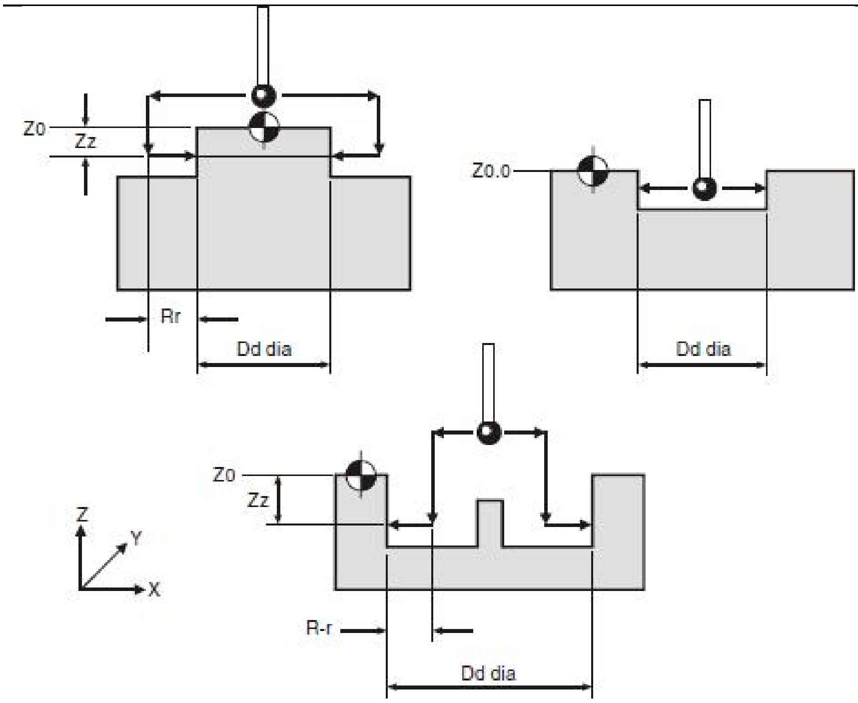
09821:Single plane measurement with angle;
09822:Measurement of angled protrusions or grooves
09823:3-point inner hole or outer circle measurement

d. Additional Loop

09817:Measure along the fourth axis of the Y distribution of measurement points (using the same method as 09817)
09818:Measure along the fourth axis of the Y distribution of measurement points(The method is the same as above09817)
09819:Inner hole/outer circle measurement on PCD
09820:Measurement of blank allowance;

2. Specific measurement methods:

a. Measurement of inner hole/outer circle(09814):



G65P9814 Dd or G65P9814 Dd Zz

Note: Diameter of D-shaped surface; Z: Measure the absolute position of the Z-axis when measuring holes with external or internal obstacles,
If Z is omitted, it will be assumed to be an inner hole cycle. When Rr is written as a negative number, it indicates the presence of obstacles inside

Example 2 Measure an inner hole (relative to the reference)

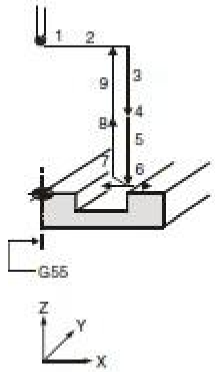
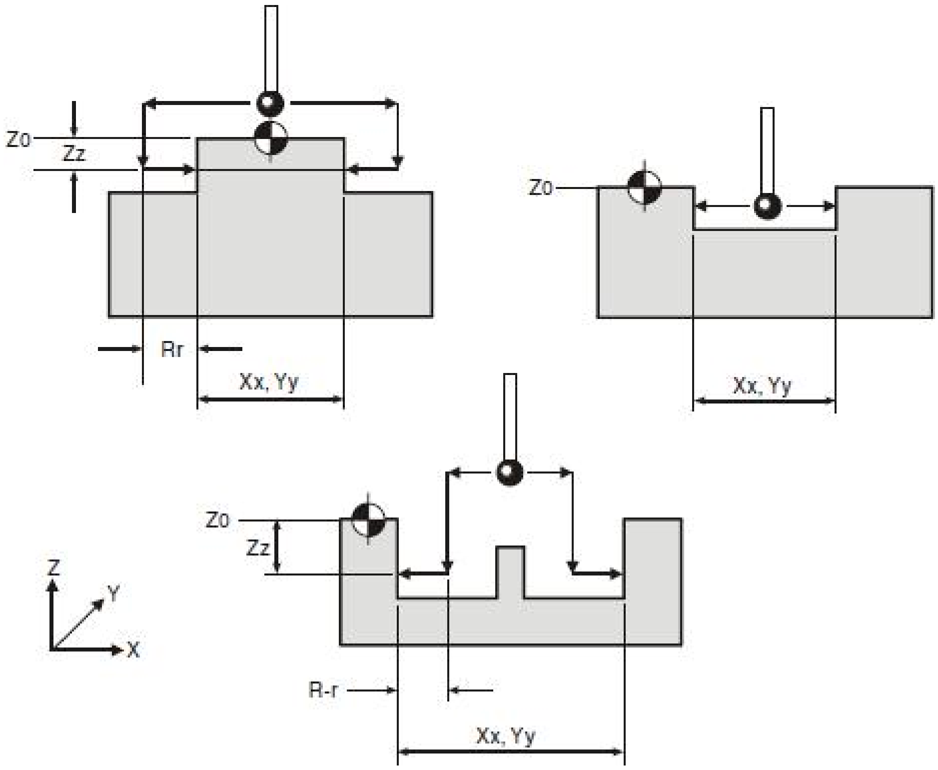


Figure 7.5
Probe movements

1.	T01M06	Select the probe.
2.	G54X100.Y50.	Start position.
3.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in).
4.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z-10.F3000.	Protected positioning move.
6.	G65P9812X30.S2	Measure a 30.0 mm (1.181 in) wide pocket.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (where applicable).
9.	G28Z100.	Reference return.

continue

b Measurement of protrusions/grooves (09812)



Explain

This program is used to measure inner holes or outer circles, and it uses four measurement movements along the X and Y axes.

Application

With the probe and probe bias effective, position the probe on the centerline of the profile and at the appropriate Z-axis position, and run this loop with the appropriate input content according to the instructions

format:

G65p9812Xx or
G65P9812Yy or
G65P9812Xx Zz
or
G65P9812Yy Zz

Note: X: Measure distance when measuring along the X direction;

Y: Measure distance when measuring along the Y direction;:Z:
Measure the absolute position of Z when measuring a convex

Example 1 Convex measurement

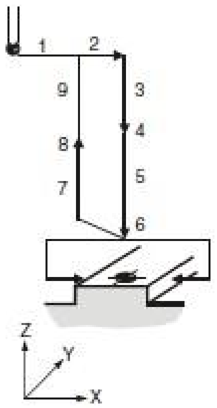


Figure 7.4
Probe movements

1.	T01M06	Select the probe.
2.	G54X0Y0	Start position.
3.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in) above.
4.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z10.F3000	Protected positioning move.
6.	G65P9812X50.Z-10.S2	Measure a 50.0 mm (1.968 in) wide web.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (where applicable).
9.	G28Z100.	Reference return.

continue

platform or a groove with obstacles inside. If Z is missing, it means measuring a groove;

Example 2 Groove measurement (relative reference)

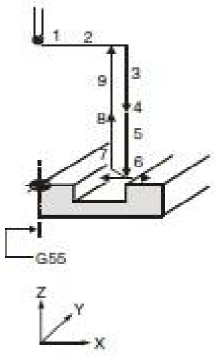
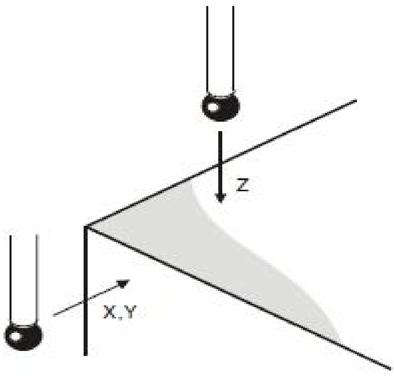


Figure 7.5
Probe movements

1.	T01M06	Select the probe.
2.	G54X100.Y50.	Start position.
3.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in).
4.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z-10.F3000.	Protected positioning move.
6.	G65P9812X30.S2	Measure a 30.0 mm (1.181 in) wide pocket.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (where applicable).
9.	G28Z100.	Reference return.

continue

c Measurement of a single plane (09811)



Explain

This cycle measures a plane and its size or position has been determined

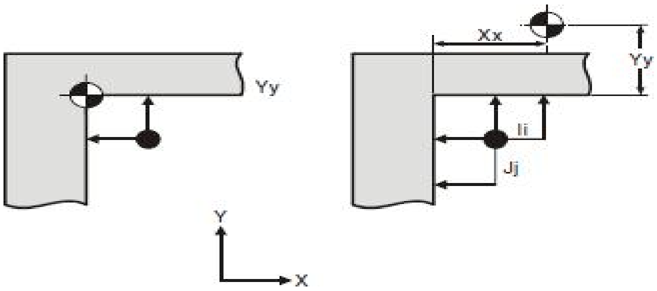
Application

Position the measuring head close to the surface with the effective deviation of the probe blade, perform a cyclic measurement of the surface, and return to the starting position after the measurement is completed

There are two possibilities:

1. The plane can be viewed as a tool deviation change corresponding to the dimensions of Tt and Hh;

2. The plane can be regarded as a reference plane position, and the coordinate system can be adjusted through Ss and Mm;d.(09815)Inner corner measurement;



Explain

This loop is used to determine the position of the corner. Note that even if the corner is not 90 degrees, the actual intersection point of the corner can still be found

Application

In the case of effective deviation of the measuring head blade, the measuring head must be positioned at the starting position shown in the above figure. The measuring head first measures the Y plane, then measures the X plane, and then returns to the starting position.

If a malfunction occurs during the cycling process, the probe will return to its starting position

Note: If I and J are omitted, only two measurement movements will occur, and it is assumed that the corner is parallel to the coordinate axis; If I or J is omitted, then three measurement movements occur and it is assumed that the corner is 90 degrees;

Format: G65 P9815Xx Yy;

Output the following content:

1. Corner position
2. Tolerance (if used)
3. Deviation in the X-axis direction
4. Deviation in the Y-axis direction
5. Zero offset number (if used)

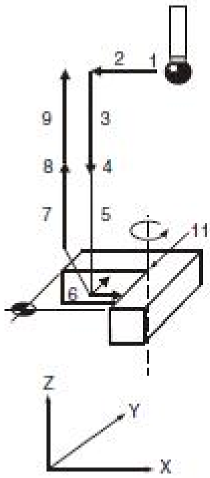


Figure 7.10
Probe movements

1.	T01M06	Select the probe.
2.	G54X10.0Y10.	Start position.
3.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in).
4.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z-5.F3000	Protected positioning move.
6.	G65P9815X20.Y20.I10.J10.	Corner find.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (where applicable)
9.	G28Z100.	Reference return.
10.	G17	Select plane.
11.	G68X#135Y#136R#139	Set rotational position and angle.
	continue machining	
12.	G69	Cancel rotation mode.

e. (O9816)Measurement of outer corners;

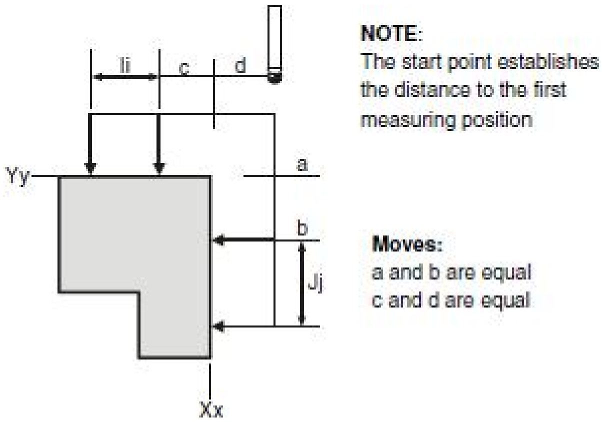


Figure 7.11 Outer corner measurement

Explain the movement

a=b

c=d

This loop is used to determine the position of the corner. Note that even if the corner is not 90 degrees, the actual intersection point of the corner can still be found.

Application

In the case of effective deviation of the measuring head blade, the measuring head must be positioned at the starting position shown in the figure, and the measuring head should first measure the Y plane. Then measure the X plane, and the probe returns to the starting value.

If a malfunction occurs during the cycling process, the probe will return to its starting position

format:G65 P9816 Xx Yy

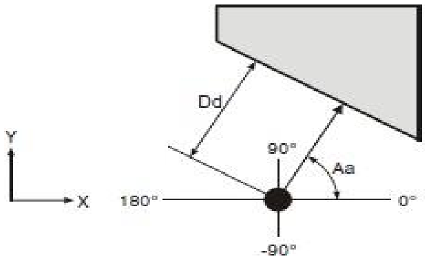
X The nominal position of the X-axis at the corner

Y The nominal position of the Y-axis at the corner

Output the following content:

1. Corner position
2. Tolerance (if used)
3. Deviation in the X-axis direction
4. Deviation in the Y-axis direction
5. Zero offset number (if used)

f.(O9821)Single plane measurement with angle



NOTE:
Angles in the range ±180 degrees
+Aa angles (counter-clockwise)

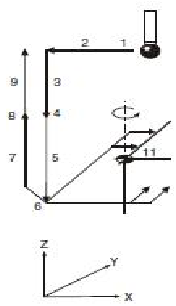


Figure 7.12
Probe movements

1.	T01M06	Select the probe.
2.	G54X-10.Y-10.	Start position.
3.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in).
4.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z-5.F3000	Protected positioning move.
6.	G65P9816X0Y0I10.J10.	Corner find.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (where applicable).
9.	G28Z100.	Reference return.
10.	G17	Select plane.
11.	G68X#135Y#136R#139	Set corner position and angle.
	continue machining	
12.	G69	Cancel co-ordinate rotation mode.

Explain

This loop utilizes vector measurement movement along the XY axis to measure the features of a plane

Application

With the probe and probe bias effective, place the probe at the desired reference position on the profile and position it at the appropriate Z-axis position. Follow the instructions to run this loop with the appropriate input content

Format：G65 P9821 Aa

a When measuring from the X+direction, the measuring direction of the probe.

B Nominal distance to surface (radial)

Dd

Example: Angled single surface measurement

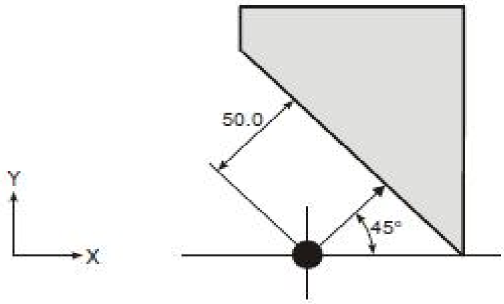


Figure 8.2 Measuring a single angled surface

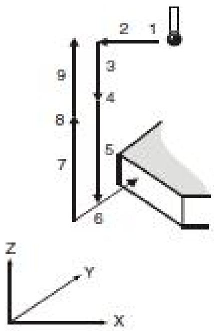


Figure 8.3
Probe movements

1.	T01M06	Select the probe.
2.	G54X-40.Y20.	Start position.
3.	G43H1Z100.	Activate offset 1, go to 100 mm (3.94 in).
4.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z-8.F3000	Protected positioning move to start position.
6.	G65P9821A45.D50.T10	Single surface measure.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (when applicable).
9.	G28Z100.	Reference return.

g.（O9822）Measurement of angled protrusions or grooves

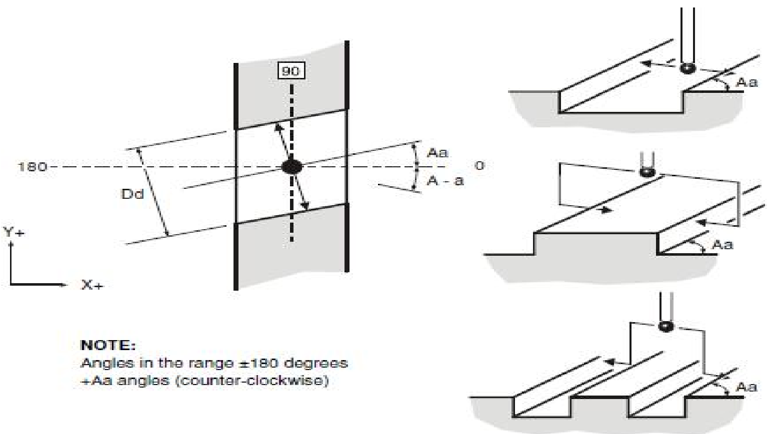


Figure 8.4 Measuring an angled web or pocket

Explain

This cycle utilizes two vectors along the X and Y axes to measure the movement of the convex platform or groove

Application

With the probe and probe bias effective, position the probe on the centerline of the profile and at the appropriate Z-axis position, and run this loop with the appropriate input content according to the instructions

format：G65 P9822 Aa Dd

a The angle calculated from the X-axis direction of the side plane

b Nominal size of the profile

When measuring the absolute position of the Z-axis on the convex platform, if this parameter is omitted, it will be set as a groove cycle

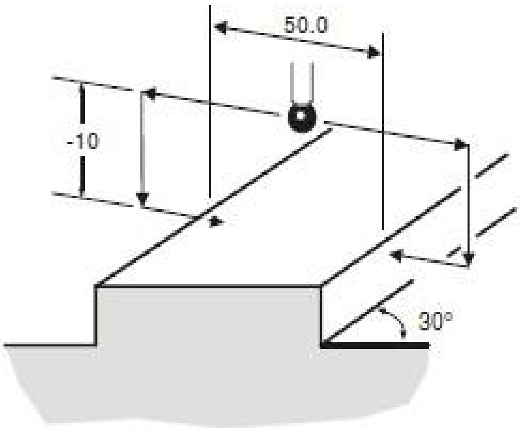


Figure 8.5 Measuring an angled web

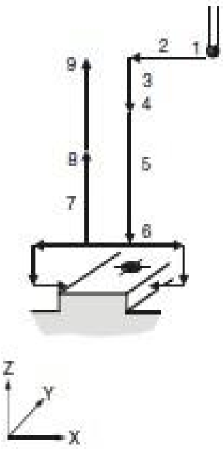


Figure 8.6
Probe movements

1.	T01M06	Select the probe.
2.	G54X0Y0	Start position.
3.	G43H1Z100.0	Activate offset 1, go to 100 mm (3.94 in).
4.	G65 P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z10.F3000	Protected positioning move.
6.	G65P9822A30.D50.Z-10.S2	Measure a 50.0 mm (1.9685 in) wide web at 30 degrees.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (when applicable).
9.	G28Z100.	Reference return

continue

The feature centre line in the X-axis is stored in the work offset S02 (G55).

h. （09823）3-point inner hole or outer circle measurement

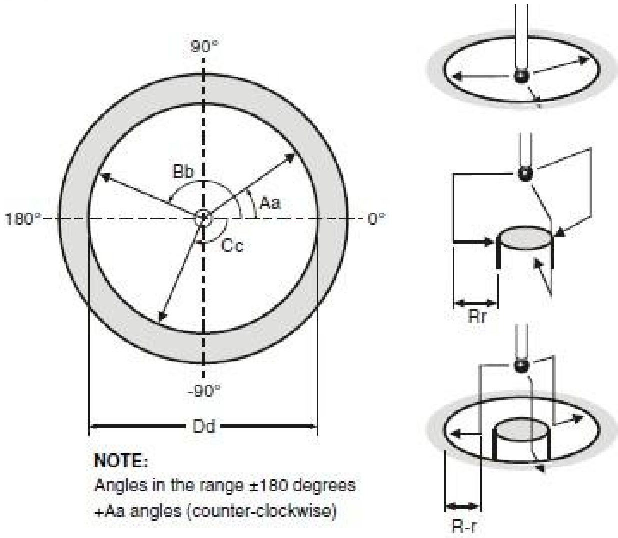


Figure 8.7 3-point bore or boss measurement

Explain

This program uses three vector measurements along the XY axis to move and measure the inner hole/outer circle

Application

When the probe and probe bias are effective, position the probe on the neutral line of the profile and in the appropriate Z-axis position, and run this loop with the appropriate input content according to the instructions

Format:G65 P9823 Aa Bb Cc Dd OR G65 P9823 Aa Bb Cc Dd Zz

a The angle of the first vector measurement is calculated from the X+direction

b The angle of the second vector measurement is calculated from the X+direction

C The angle of the third vector measurement is calculated from the X+direction

D Nominal size of the profile

When measuring the outer circle, the absolute coordinate position of the Z-axis. If this parameter is omitted, it will falsely locate an inner hole cycle

h. （09817）Measurement points are measured along the fourth axis of the

Example: 3-point bore measurement (referred datum)

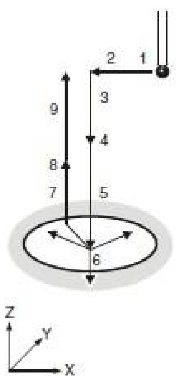


Figure 8.8
Probe movements

1.	T01M06	Select the probe
2.	G54X100.Y100.	Start position
3.	G43H1Z100.	Activate offset 1 go to 100 mm (3.94 in)
4.	G65 P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
5.	G65P9810Z-10.F3000	Protected positioning.
6.	G65P9823D30.A30.B150.C-90.S2	Measure a 30.0 mm (1.181 in) diameter bore.
7.	G65P9810Z100.	Protected positioning move.
8.	G65P9833	Spin the probe off (when applicable).
9.	G28Z100.	Reference return

continue

The error of centre line is referred to the datum point X0,Y0 and the revised X0,Y0 position is set in work offset 02 (G55).

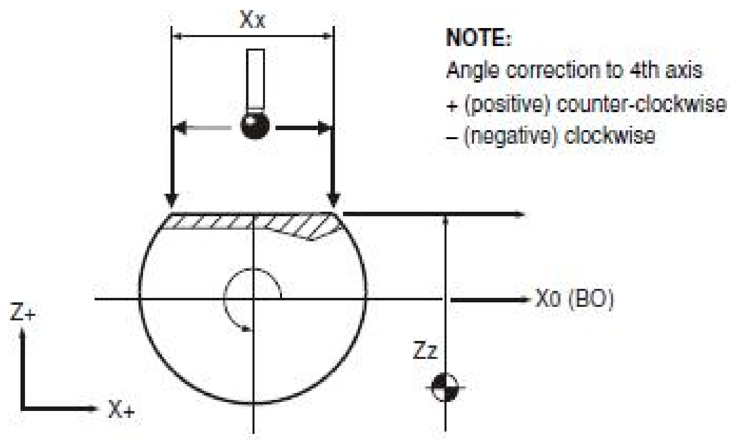


Figure 9.1 4th axis X measurement

X distribution

Explain
The purpose of this subroutine is to find a plane tilt angle between two points (Z1 and Z2) and compensate for surface errors by rotating the fourth axis
Application
The fourth axis must be positioned at the desired angular position on the profile - that is, the surface is perpendicular to the Z-axis. If R19=s input is used, the zero bias register will be adjusted based on this deviation
Note: For the vast majority of machine tools, it is necessary to reassign the zero offset and move to this angular position after this cycle in order to activate the new zero offset.

Coordinate system setting S

X The X-direction distance between measurement points Z1 and Z2
z Expected surface position of Z-axis

Outputs

#143 will show (Z1 - Z2) value

#144 will show angle correction value

#139 will show 4th axis measured position.

NOTE: Different machines and application may require the 4th axis system variable number to be changed. It will be achieved by editing macro O9817 when the macro is installed to suit your machine.

Edit as follows:

#3 = 4 (4th axis number) change axis number as required (see the Fanuc macro section for axis numbers)

Axis direction change

Edit as follows:

#4 = 1(1 = clockwise, and -1 = counter-clockwise) change as required.

Example: Set the 4th axis to a milled flat

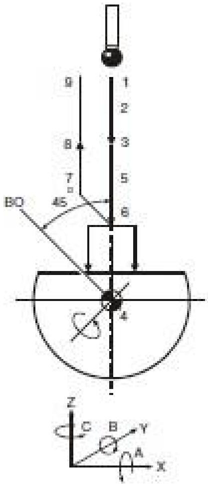
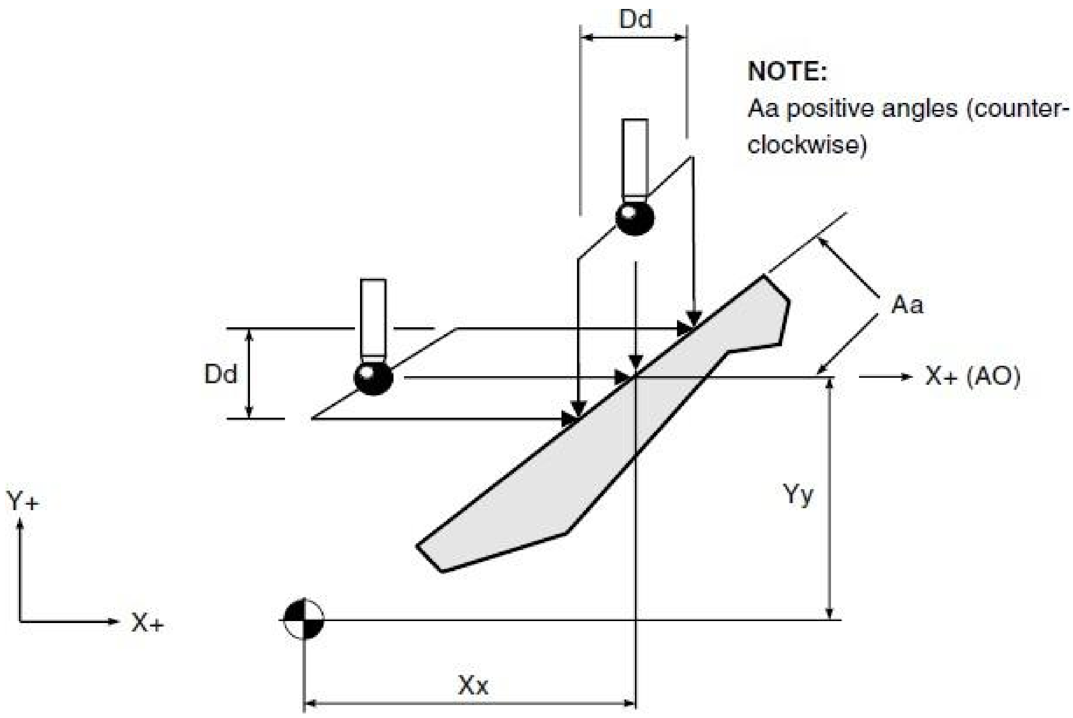


Figure 9.2
Probe movements

1.	T01M06	Select the probe.
2.	G43H1Z200.	Activate offset 1, go to 100 mm (3.94 in).
3.	G65P9832	Spin the probe on (includes M19), or M19 for spindle orientation.
4.	G0B45.	Position 4th axis to 45 degrees.
5.	G65P9810X0Y0Z20.F3000	Position 10 mm (0.394 in) above the surface.
6.	G65P9817X50.Z10.B5.	Measure at 50 mm (1.9685 in) centres, update G54 and set a tolerance of 5 degrees.
7.	G65P9810Z200.	Protected positioning move.
8.	G65P99833	Spin the probe off (when applicable).
9.	G28Z200.	Reference return.

continue

i. (09843) Angle measurement of X or Y plane



Explain

This cycle takes X or Y measurements at two positions to determine the angular position of the surface

Application

In order to provide an appropriate starting position, the probe must be positioned on the surface attachment and in the required Z-axis position. The cycle will produce two measurements of symmetry and starting position to determine the surface angle

format: G65 P9843 Xx Dd or G65P9843Yy Dd

remarks: X The midpoint position of the surface

y The midpoint position of the surface

d The movement distance parallel to the X or Y axis between two measurement positions

Example

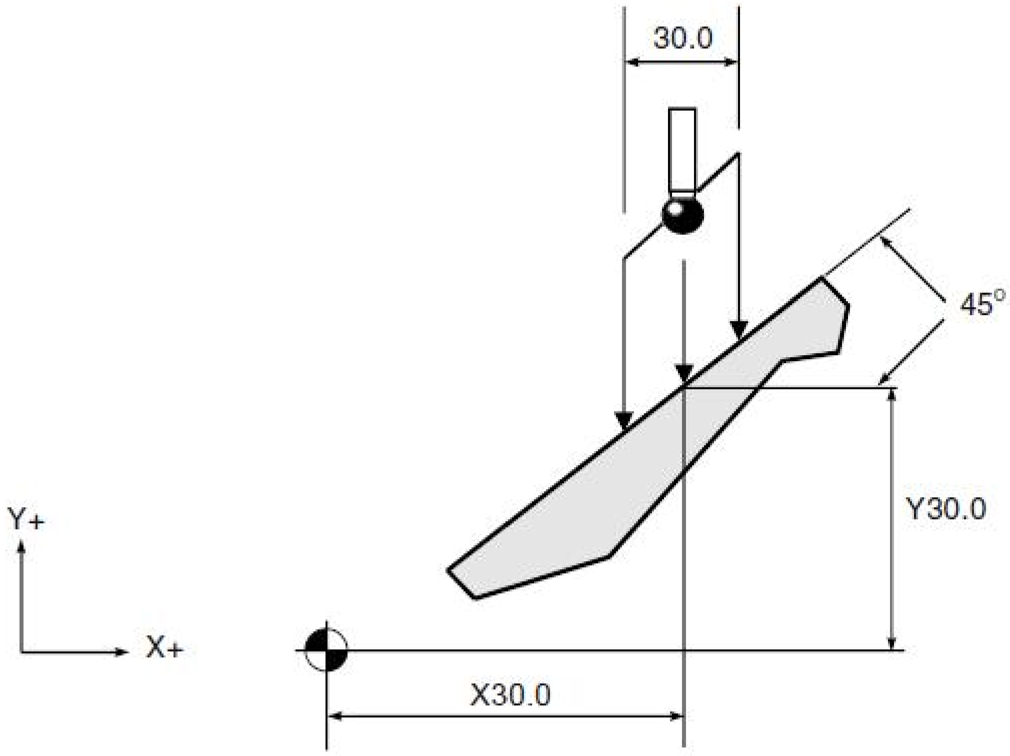


Figure 9.21 Example of an angled surface measurement

G65P9810 X30.Y50.Z100.F3000	Protected positioning move.
G65P9810 Z-15.	Protected move to start position.
G65P9843Y30.D30.A45.	Angle measure.
G65P9810 Z100.	Retract to a safe position.
continue	
G17	
G68G90X0Y0 R[#139]	Rotate co-ordinate system by the angle.

j. （09834）:Measure data from the surface to the surface within the XY plane

:Measure data from profile to profile between Z-planes

(1) 09834 :Measure data from the surface to the surface within the XY plane

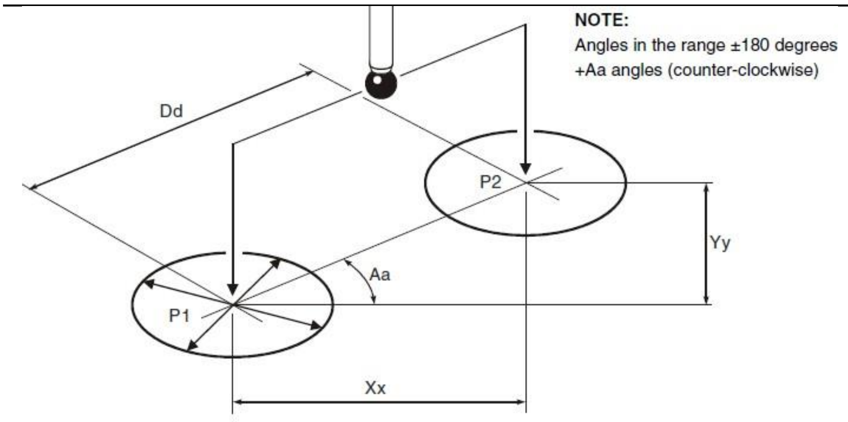


Figure 9.12 Determining feature-to-feature data in the XY plane

Note:

Note: The order of P1 and P2 is very important because the calculated P2 data is relative to P1.

The value of P1 is obtained by calling L9834 (without any input content) after the first measurement cycle.

The value of P2 was obtained by running the second measurement cycle, and the values between the profiles were obtained by adding appropriate inputs and calling L9834 after the second cycle

09834

format:

G65 P9834 Xx或G65 P9834 Yy或G65 P9834 Xx Yy或G65 P9834 Aa Dd或G65 P9834

remarks:

X Nominal incremental distance in the X-axis direction

Y Nominal incremental distance in the Y-axis direction

a Starting from the X-axis direction, calculate the angle of P2 relative to P1 (within ± 180 degrees)

b The shortest distance between P1 and P2

Example 1 Measuring the incremental distance between two ho

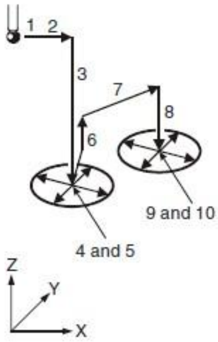


Figure 9.13
Probe movements

1. G65P9810X30.Y50.F3000 Protected positioning move.
2. G65P9810Z-10. Protected positioning move.
3. G65P9814D20. P1 20 mm (0.787 in) bore.
4. G65P9834 Store data.
5. G65P9810Z10. Protected positioning move.
6. G65P9810X80.Y78.867 Move to new position.
7. G65P9810Z-10. Protected positioning move.
8. G65P9814D30. P2 30 mm (1.181 in) bore.

And either this

9. G65P9834X50.Y28.867M.1 Incremental distance measure with 0.1 mm (0.0039 in) true position tolerance.

or this

9. G65P9834A30.D57.735M.1

Example 2: Surface to bore measurement

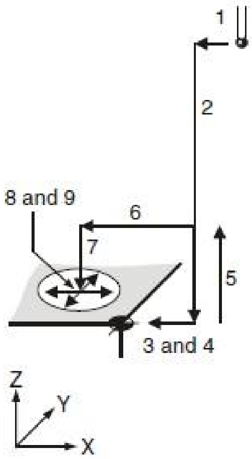
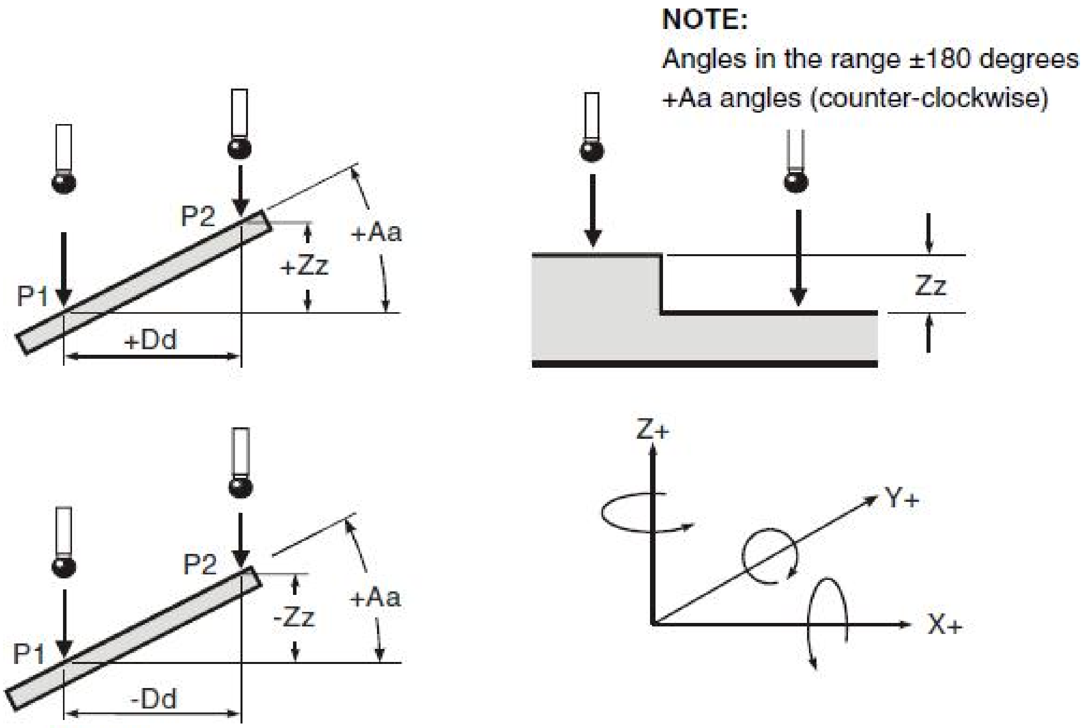


Figure 9.14
Probe movements

1. G65P9810X10.Y50.F3000 Protected positioning move.
2. G65P9810Z-10. Protected positioning move.
3. G65P9811X0. P1 at X0 mm (0 in) position.
4. G65P9834 Store data.
5. G65P9810Z10. Protected positioning move.
6. G65P9810X-50. Move to new position.
7. G65P9810Z-10. Protected positioning move.
8. G65P9814D20.5 P2 20.5 mm (0.807 in) bore.
9. G65P9834X-50.H.2 Measure distance -50 mm (-1.97 in)

(2) O9834 : Measure data from profile to profile between Z-planes



Explain

This is a non moving subroutine used after two measurement cycles to determine data between profiles

Note: The order of P1 and P2 is very important because the calculated P2 data is relative to P1.

The value of P1 is obtained by calling L9834 (without any input content) after the first measurement cycle.

The value of P2 was obtained by running the second measurement cycle, and the values between the profiles were obtained by adding appropriate inputs and calling L9834 after the second measurement cycle

format: G65 P9834Zz or

G65 P9834Aa Zz or G65 P9834Dd Zz or G65 P9834

Explain:

1. Updating a tool offset with T input is only possible if O9811 is used for P2 data. Otherwise an alarm (T INPUT NOT ALLOWED) results.

2. Angles. These are with respect to the XY. Use angles in the range ±180 degrees.

3. When G65P9834 (without any inputs) is used, then the following data is stored:

from	#135	to	#130
	#136		#131
	#137		#132
	#138		#133
	#139		#134

Inputs

Aa Zz or Dd Zz inputs

1. +Dd/-Dd values should be used to indicate the direction of P2 with respect to P1.

2. +Aa angles (counter-clockwise).

3. Angles between ±180 degrees.

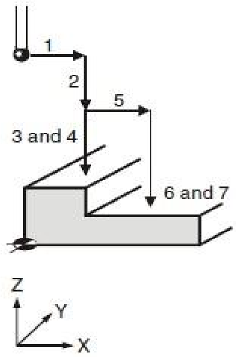
Zz only input

+Zz/-Zz values should be used to indicate the direction of P2 with respect to P1.

Required input section：

Aa	a=	The angle of P2 with respect to P1 measured from the XY plane (angles between ±180 degrees).
Zz	z=	The nominal incremental distance in the Z axis.
or		
Dd	d=	The minimum distance between P1 and P2 measured in the XY plane.
Zz	z=	The nominal incremental distance in the Z axis.

Example 1: Measuring the incremental distance between two surfaces



1.	G65P9810X30.Y50.F3000	Protected positioning move.
2.	G65P9810Z30.	Protected positioning move.
3.	G65P9811Z20.	P1 20 mm (0.787 in) surface.
4.	G65P9834	Store data.
5.	G65P9810X50.	Move to new position.
6.	G65P9811Z15.	P2 15 mm (0.591 in) surface.
7.	G65P9834Z-5.H.1	Feature to feature at -5.0 mm (-0.197 in).

Figure 9.16
Probe movements

Example 2: Measuring an angled surface

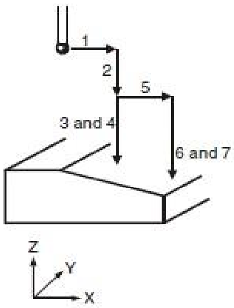


Figure 9.17
Probe movements

1. G65P9810X30.Y50.F3000 — Protected positioning move.
2. G65P9810Z30. — Protected positioning move.
3. G65P9811Z20. — P1 at 20 mm (0.787 in) position.
4. G65P9834 — Store data.
5. G65P9810X77.474 — Move to new position.
6. G65P9811Z10. — P2 at 10 mm (0.394 in) position.

and either this

7. G65P9834D27.474Z-10.B.5 — Measure slope -20 degrees (clockwise) angle tolerance ±5 degrees.

or this

7. G65P9834A-20.Z-10.B.5

4. The meanings represented by each macro variable

Output Variables - Table 1

	Single plane	Convex/groove	Inner hole/outer circle	Inner corner	Outer corner	Fourth axis	XY plane angle
	G65P9811	G65P9812	G65P9814	G65P9815	G65P9816	G65P9817/18	G65P9845
#135	X position	X position	X position	X position	X position		
#136	Y position	Y position	Y position	Y position	Y position		
#137	Z position						
#138	size	size	size				
#139				X-plane angle	X-plane angle	Fourth axis angle	angle
#140	X deviation	X deviation	X deviation	X deviation	X deviation		
#141	Y deviation	Y deviation	Y deviation	Y deviation	Y deviation		
#142	Z deviation			Y-plane angle	Y-plane angle		
#143	Dimensional deviation	Dimensional deviation	Dimensional deviation	Y angle deviation	Y angle deviation	Height deviation	Height deviation
#144				X angle deviation	X angle deviation	Angle deviation	Angle deviation

#145	Actual location deviation	Actual location deviation	Actual location deviation	Actual location deviation	Actual location deviatio n		
#146	Remaining condition	Remaining condition	Remaining condition				
#147	Direction indicator						
#148	Exceeding the tolerance mark (1 to 7)						
#149	Probe error flag (0 to 2)						

Output Variables - Table 2

	PCD inner hole/out er circular	blank stock	Flat with angles noodles	Convex with angles Platform /groove	3-point inner hole/ou ter hole Circle measure ment	From profile to profile
	G65P9819	G65P9820	G65P9821	G65P9822	G65P9823	G65P9817/ 34
#135	X position		X from the starting point position	X position	X position	Incremental distance in X direction
#136	Y position		Y from the starting point position	Y position	Y position	Incremental distance in Y direction
#137	PCD					Incremental distance in Z direction
#138	size		The ruler starting from the starting point inch	size	size	minimum distance
#139	angle					angle
#140	X deviatio n		X deviation	X deviatio n	X deviatio n	X deviation
#141	Y deviatio		Y deviation	Y deviatio	Y deviatio	Y deviation

	n			n	n	
#142	PCD deviation					Z deviation
#143	Dimensional deviation		Dimensional deviation	Dimensional deviation	Dimensional deviation	Minimum distance deviation
#144	Angle deviation	Maximum value				Angle deviation
#145	Actual positional deviation	minimum value	Actual positional deviation	Actual positional deviation	Actual positional deviation	Actual positional deviation
#146	Remaining condition	Variable momentum (blank)	Remaining condition	Remaining condition	Remaining condition	Remaining condition
#147	Hole number		Direction indicator			
#148	Exceeding the tolerance mark (1 to 7)					
#149	Probe error flag (0 to 2)					

Common fault handling

Equipment	Abnormal	Reason	Handling method
Probe	Red light flashing	Weak signal	Adjust the installation distance and angle of the receiver
Probe	Green light on when not triggered	internal fault	Return to factory
Probe	The white light is on	Low battery voltage	Replace battery
Receiver	The receiver power light is not on	Power not input	Check the wiring Confirm that the wiring is intact but still does not light up, return to the factory
Receiver	The white light is on	Low voltage of probe battery	Replace battery
Receiver	Red light flashing	Weak signal	Adjust the installation distance and angle of the receiver

Product List

1 wireless probe
1 ruby measuring needle (0.001 roundness)
1 signal receiver